

ARTIFICIAL INTELLIGENCE (AI)

Milestone 3 - Final Report

Automatic Temporal Human Action Detection Using a CLIP-Based Video Recognition System with a Gradio User Interface

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Project summary: In Milestone 2, we used a pretrained CLIP model to recognize actions from selected video segments. In Milestone 3, we extended that prototype so the system scans a complete multi-action video automatically, predicts the action in overlapping time windows, and displays the result through a Gradio interface. The final tested video contains four actions: swimming, riding a motorcycle, playing guitar, and playing basketball.

Core modification	Automatic temporal action detection
UI addition	Upload video + Submit + view results and timeline
Window settings	3-second windows, 2-second step, 3 sampled frames/window
Primary test	33.3 seconds, 60 FPS, 17 windows
Final detected sequence	Swimming -> Motorcycle -> Guitar -> Basketball

1. Background and Objective

Human action recognition asks an AI system to determine what a person is doing in a video. For a continuous video that contains several activities, a useful system must identify both the action and the approximate time at which it occurs.

The Milestone 3 objective was to convert the Milestone 2 prototype from a manually selected segment classifier into an **automatic temporal action-detection system**. A second goal was usability: a user should be able to upload a new video and run the complete pipeline without manually executing several notebook cells.

2. Research Foundation

The project is inspired by the video-text matching idea of **ActionCLIP**. Our implementation uses a pretrained **CLIP** model on sampled video frames and compares the visual content with natural-language descriptions of candidate actions.

Academic accuracy note: We do not claim to reproduce the full ActionCLIP training architecture. Instead, we built a lightweight CLIP-based action-recognition prototype inspired by the ActionCLIP video-text matching idea, and added automatic temporal detection as our Milestone 3 contribution.

ActionCLIP paper	https://arxiv.org/abs/2109.08472
CLIP paper	https://arxiv.org/abs/2103.00020
ActionCLIP implementation	https://github.com/sallymmx/ActionCLIP

3. What Changed in Milestone 3

Milestone 2	Milestone 3
User manually selected action segments.	Program scans the whole video automatically.
Example: swimming = 0-7 s.	Program creates overlapping time windows itself.
Classify selected segments.	Classify every window and build a timeline.
Several notebook cells were needed for a new test.	Gradio UI provides upload-and-analyze workflow.

4. System Pipeline

Step	What the system does
1	User uploads a video through the Gradio interface.
2	The program reads video duration and FPS.
3	The full video is divided into overlapping 3-second windows with a 2-second step.
4	Three representative frames are sampled from each window.
5	CLIP compares each frame with candidate action prompts.
6	Frame scores are averaged for each candidate action.
7	The highest-scoring action becomes the prediction for that time window.
8	Predictions are shown as text and as a temporal timeline graph.

5. User Interface Implementation

A Gradio interface was added inside Google Colab. The interface hides the repetitive notebook steps from the user. A user can select a video, click **Submit**, and receive the complete automatic analysis, including the detected actions and the timeline graph.

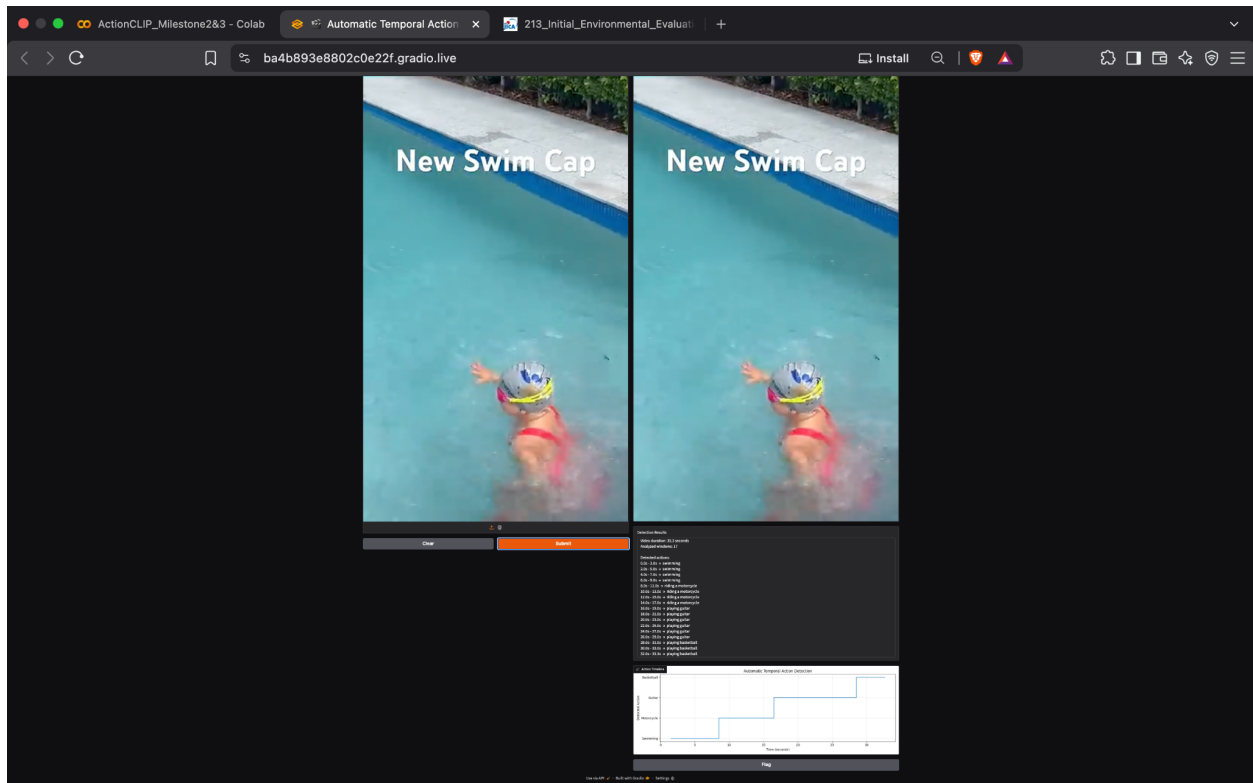


Figure 1. Working Gradio interface after analyzing the final four-action test video. The output panel shows basketball for the final three windows.

Component	Purpose
Upload Video	Select a test video.
Submit	Run the full temporal action-detection pipeline.
Input Video	Preview the uploaded video.
Detection Results	Show one action prediction for each time window.
Action Timeline	Visualize how the predicted action changes over time.

6. Main Code Logic

```
# Automatic temporal windows
window_size = 3
step_size = 2
windows = []

start = 0.0
while start < duration:
    end = min(start + window_size, duration)
    windows.append((start, end))
    start += step_size

# Candidate actions
CANDIDATE_ACTIONS = [
    "swimming",
    "riding a motorcycle",
    "playing guitar",
    "playing basketball"
]
```

The Gradio callback wraps the full analysis pipeline into one function, so a new video can be tested without manually rerunning the individual processing cells.

7. Final Test Results

The final tested video is 33.3 seconds long at 60 FPS. The system automatically created **17 overlapping windows**. Unlike the first development run, the final candidate-action set included four actions: swimming, riding a motorcycle, playing guitar, and playing basketball.

Window	Detected action
0.0-3.0 s	Swimming
2.0-5.0 s	Swimming
4.0-7.0 s	Swimming
6.0-9.0 s	Swimming
8.0-11.0 s	Riding a motorcycle
10.0-13.0 s	Riding a motorcycle
12.0-15.0 s	Riding a motorcycle
14.0-17.0 s	Riding a motorcycle
16.0-19.0 s	Playing guitar
18.0-21.0 s	Playing guitar
20.0-23.0 s	Playing guitar
22.0-25.0 s	Playing guitar
24.0-27.0 s	Playing guitar
26.0-29.0 s	Playing guitar
28.0-31.0 s	Playing basketball
30.0-33.0 s	Playing basketball
32.0-33.3 s	Playing basketball

7.1 Final Temporal Sequence

Approximate interval	Detected action
0-9 s	Swimming
8-17 s	Riding a motorcycle
16-29 s	Playing guitar
28-33.3 s	Playing basketball

Key result: The updated final run detected all four visible action stages in the tested video: **Swimming -> Riding a Motorcycle -> Playing Guitar -> Playing Basketball**. The final three windows were correctly classified as playing basketball, resolving the earlier three-action candidate-list limitation.

8. Limitations

1	Fixed windows can overlap action transitions.
2	The candidate action vocabulary is currently predefined.
3	The experiment is a small controlled prototype test, not a large benchmark evaluation.
4	The model uses sampled frames rather than an end-to-end temporal video encoder.

9. Future Improvements

Future work can add an **Unknown/Transition** class, adaptive window sizes, more sampled frames, a larger action vocabulary, and stronger temporal video models. Evaluation on a standard action dataset such as UCF101 would also provide a more rigorous benchmark.

10. Conclusion

Milestone 3 extends the Milestone 2 prototype in two practical ways. First, the system automatically scans a complete multi-action video using overlapping windows and predicts the action over time. Second, the Gradio interface makes the pipeline easy to test: a user uploads a video and clicks Submit to receive the detection results and timeline. In the final tested video, the system detected the four-action sequence swimming, motorcycle riding, guitar playing, and basketball playing.

11. Project Resources

Tested video	https://drive.google.com/file/d/1OSVEv546_Xg6cUXi4kPUdKEEzkuqr_Ka/view?usp=sharing
Project code / UI	https://colab.research.google.com/drive/1nkFcvG8azvQXDNqAknv6GrlcoJCinCIJ?usp=sharing
ActionCLIP paper	https://arxiv.org/abs/2109.08472
CLIP paper	https://arxiv.org/abs/2103.00020
ActionCLIP GitHub	https://github.com/sallymmx/ActionCLIP

Final accuracy note: The report records the actual observed UI output. It does not claim a benchmark-level accuracy score. The experiment demonstrates correct temporal identification of the four action stages in this controlled test video.